Week 3: Number Theory and Modulus

Prime Factorization

According to the *Fundamental Theorem of Arithmetic*, each integer greater than 1 can be represented either as a prime or a product of primes, and that this representation/factorization is unique. Succintly,

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} = \prod_{i=1}^{k} p_i^{e_i}$$

We consider the number of factors of n. Note that each factor can be expressed as a product of primes $p_1, p_2, ..., p_k$ raised to exponents that are at most those in the factorization of n. We have $e_i + 1$ choices (0 to e_i) for the exponent for p_i, which means the number of factors is equal to $\boxed{(e_1+1)(e_2+1)\cdots(e_k+1)}$. For instance, $20 = 2^2 * 5^1$ has $(2+1)(1+1) = 6$ factors.

Modulus

As definition, we say that $n \equiv a \pmod{m}$ for $0 \leq a < m$ when n divided by m leaves a remainder of a. For example, $2 \equiv 14 \pmod{12}$, which is similar to how 2 o'clock and 14 o'clock occupy the same position on a traditional 12 hour clock. In general, we have $a \equiv b \pmod{n}$ whenever $a - b$ is divisible by n.

Suppose that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Here a few useful properties.

$$\begin{aligned} a + c &\equiv b + d \pmod{n} \\ a - c &\equiv b - d \pmod{n} \\ ac &\equiv bd \pmod{n} \\ \frac{a}{e} \equiv \frac{b}{e} \pmod{\frac{n}{\gcd(n,e)}} &\quad \text{for } e \text{ is a positive integer} \\ a^e \equiv b^e \pmod{n} &\quad \text{for } e \text{ is a positive integer} \end{aligned}$$

Let p be a prime and a be any positive integer. According to *Fermat's Little Theorem*, then $a^p - a$ is divisible by p, which is equivalent to $a^p \equiv a \pmod{p}$, which dissolves into

$$a^{p-1} \equiv p \pmod{p}.$$

Euler's Theorem

A generalized form of this is *Euler's Theorem*, which states that if a and n are coprime positive integers, then $a^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n)$ is Euler's totient function. This represents the number of positive integers less than n that are relatively prime to n. Suppose that n has prime factors $p_1, p_2, ..., p_k$. Then,

$$\phi(n) = n(1 - \frac{1}{p_1})(1 - \frac{1}{p_2})\cdots(1 - \frac{1}{p_k}) = n\prod_{i=1}^{k}(1 - \frac{1}{p_i}).$$

For instance, $\phi(10) = 10(1 - \frac{1}{2})(1 - \frac{1}{5}) = 4$. This would mean that $a^4 \equiv 1 \pmod{10}$ for $\gcd(a, 10) = 1$.

Bases and Divisibility

The typical convention of mathematics has been to count in powers of 10, which ensures that each digit is between 0 and 9. For example, $123 = 1 * 10^2 + 2 * 10^1 + 3 * 10^0$. This is called base 10. However, there are other bases such as binary, where each digit is between 0 and 1 and ascends in powers of 2. For instance, $101_2 = 1 * 2^2 + 0 * 2 + 1 = 5_{10}$. In other words, 101 is binary is 5 in base 10. Babylonian cultures used base 60, which can be seen in the construction of measured time, with 60 seconds in a minute and 60 minutes in an hour. In base 60, the number $10_{\text{base 60}}$ would translate to $1 * 60^1 + 0 * 60^0$, or $60_{\text{base 10}}$.

In general, let a_n denote a number a written in base n. Supposing that a is a k-digit number with $a_n = \overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0}$, then converting a_n to base 10 leads to

$$\overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0} = a_{k-1} * n^{k-1} + a_{k-2} * n^{k-2} + \cdots + a_0 n^0.$$

By definition, a base 10 number can be expressed as

$$\overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0} = a_{k-1} * 10^{k-1} + a_{k-2} * 10^{k-2} + \cdots + a_0 10^0.$$

As a few samples,

$$123456_{\text{base 10}} = 1 * 10^5 + 2 * 10^4 + 3 * 10^3 + 4 * 10^2 + 5 * 10^1 + 6 * 10^0$$

Divisibility

We can utilize this digital representation of base 10 numbers to prove a crucial divisibility rule. The remainder when an integer n is divided by 9 is equal to the remainder when the sum of n's digits is divided by 9. In other words, $n \equiv S(n) \pmod 9$, where $S(n)$ signifies the sum of the digits of n. Express n as a k-digit number $a_n = \overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0}$. We note that $10^e \equiv 1^e \equiv 1 \pmod 9$. Thus,

$$\begin{aligned}
\overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0} &\equiv a_{k-1} * 10^{k-1} + a_{k-2} * 10^{k-2} + \cdots + a_0 10^0 \pmod 9 \\
&\equiv a_{k-1} + a_{k-2} + \cdots + a_0 \pmod 9 \\
&\equiv S(n) \pmod 9.
\end{aligned}$$

As a corollary, if $S(n)$ is divisible by 9, then so is the integer n itself. We can also use modular arithmetic to prove the divisibility rule for 11, which dictates that n is divisible by 11 *if and only* if the alternating sum of its digits is divisible by 11. Note that $10^e = (11-1)^e \equiv (-1)^e \pmod{11}$. This is equal to 1 when the exponent is even and -1 otherwise. Hence,

$$\begin{aligned}
\overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0} &\equiv \mathrm{a}_{k-1} * 10^{k-1} + a_{k-2} * 10^{k-2} + \cdots + a_0 10^0 \pmod{11} \\
&\equiv a_{k-1} * (-1)^{k-1} + a_{k-2} * (-1)^{k-2} + \cdots + a_0 (-1)^0 \pmod{11}.
\end{aligned}$$

Note that this is the sum of the alternating digits of n, and if this is equal to 0, then $n \equiv 0 \pmod{11}$. Hence, the proof is complete.

Week 3: Number Theory and Modulus Problems

Problem 1

Prove that for any six distinct positive integers, there are two of them whose absolute difference is divisible by 5.

Problem 2

What is the tens digit in the sum $7! + 8! + 9! + ... + 2006!$ (AMC 10B 2006)

Problem 3

Let $S(n)$ equal the sum of the digits of positive integer n. For example, $S(1507) = 13$. For a particular positive integer n, $S(n) = 1274$. Which of the following could be the value of $S(n+1)$? (AMC 10A 2017)

(A) 1 **(B)** 3 **(C)** 12 **(D)** 1239 **(E)** 1265

Problem 4

Find the remainder when $9 \times 99 \times 999 \times \cdots \times \underbrace{99\cdots9}_{999\text{ 9's}}$ is divided by 1000. (AIME I 2010)

Problem 5

The base-ten representation for 19! is $121, 6T5, 100, 40M, 832, H00$, where T, M, and H denote digits that are not given. What is $T + M + H$? (AMC 10B 2019)

(A) 3 **(B)** 8 **(C)** 12 **(D)** 14 **(E)** 17

Problem 6

The number n can be written in base 14 as $\underline{a}\ \underline{b}\ \underline{c}$, can be written in base 15 as $\underline{a}\ \underline{c}\ \underline{b}$, and can be written in base 6 as $\underline{a}\ \underline{c}\ \underline{a}\ \underline{c}$, where $a > 0$. Find the base-10 representation of n. (AIME I 2018)

Problem 7

The positive integers N and N^2 both end in the same sequence of four digits $abcd$ when written in base 10, where digit a is not zero. Find the three-digit number abc. (AIME I 2014)

Problem 8

Oscar buys 13 pencils and 3 erasers for 1.00. A pencil costs more than an eraser, and both items cost a whole number of cents. What is the total cost, in cents, of one pencil and one eraser? (AMC 12A 2006)

Problem 9

A palindrome between 1000 and 10, 000 is chosen at random. What is the probability that it is divisible by 7? (AMC 10B 2010)

Problem 10

Arithmetic sequences (a_n) and (b_n) have integer terms with $a_1 = b_1 = 1 < a_2 \leq b_2$ and $a_n b_n = 2010$ for some n. What is the largest possible value of n? (AMC 12A 2010)

Week 3 Class Solutions

Problem 1

Prove that for any six distinct positive integers, there are two of them whose absolute difference is divisible by 5.

Answer: Proof below.

Solution: We note that for any positive integer x, $x \pmod 5$ must be congruent ($\equiv$) to either 0, 1, 2, 3, and 4. There are five possible numbers, and since we have six distinct integers, at least two of these integers, say a and b, must be congruent modulo 5 via Pigeonhole Principle. Hence, $a \equiv b \pmod 5$, which means that $a - b \equiv 0 \pmod 5$, which culminates the proof.

Problem 2

What is the tens digit in the sum $7! + 8! + 9! + ... + 2006!$ (AMC 10B 2006)

Answer: $\boxed{4}$

Solution: We note that $x! \equiv 0 \pmod{100}$ for any $x \geq 10$ since $10!$ is divisible by 100. Thus, we only need to look at $7! + 8! + 9!$ to determine the tens digit. We note that $7! = 5040$, which means that

$$7! + 8! + 9! \equiv 7!(1 + 8 + 9 * 8) \equiv 5040(81) \equiv 40(81) \equiv 3240 \equiv 40 \pmod{100}.$$

Thus, the number has a tens digit of $\boxed{4}$, as it leaves behind a remainder of 40 is divided by 100.

Problem 3

Let $S(n)$ equal the sum of the digits of positive integer n. For example, $S(1507) = 13$. For a particular positive integer n, $S(n) = 1274$. Which of the following could be the value of $S(n+1)$? (AMC 10A 2017)

(A) 1 **(B)** 3 **(C)** 12 **(D)** 1239 **(E)** 1265

Answer: $\boxed{(D)1239}$

Solution: We argue that $S(n) \equiv n \equiv 9$. Express n as a k-digit number $a_n = \overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0}$. We note that $10^e \equiv 1^e \equiv 1 \pmod 9$ for exponent e. Thus,

$$\begin{aligned}
\overline{a_{k-1}a_{k-2}a_{k-3}\cdots a_0} &\equiv a_{k-1} * 10^{k-1} + a_{k-2} * 10^{k-2} + \cdots + a_0 10^0 \pmod 9 \\
&\equiv a_{k-1} + a_{k-2} + \cdots + a_0 \pmod 9 \\
&\equiv S(n) \pmod 9.
\end{aligned}$$

Hence, $S(n+1) \equiv n + 1 \pmod 9$ and $S(n) \equiv n \pmod 9$, which means that

$$S(n+1) - S(n) \equiv n + 1 - n \equiv 1 \pmod 9.$$

Since $S(n) \equiv 1274 \equiv 5 \pmod 9$, then $S(n+1) \equiv 6 \pmod 9$. The only answer option that satisfies this is $\boxed{(D)1239}$.

Problem 4

Find the remainder when $9 \times 99 \times 999 \times \cdots \times \underbrace{99\cdots9}_{999\text{ 9's}}$ is divided by 1000. (AIME I 2010)

Answer: $\boxed{109}$

Solution: We note that for $n \geq 3$,

$$\underbrace{99\cdots9}_{n\text{ 9's}} = 10^n - 1 \equiv 10^3 * 10^{n-3} - 1 \equiv -1 \pmod{10^3}.$$

Thus, we can write the desired expression as

$$\begin{aligned} 9 \times 99 \times 999 \times \cdots \times \underbrace{99\cdots9}_{999\text{ 9's}} &\equiv 9 * 99 * \underbrace{(-1) * (-1) * \cdots * (-1)}_{997\text{ times}} \pmod{1000} \\ &\equiv 9 * 99 * (-1)^{997} \equiv -891 \equiv \boxed{109} \pmod{1000}. \end{aligned}$$

Problem 5

The base-ten representation for 19! is $121, 6T5, 100, 40M, 832, H00$, where T, M, and H denote digits that are not given. What is $T + M + H$? (AMC 10B 2019)

(A) 3 **(B)** 8 **(C)** 12 **(D)** 14 **(E)** 17

Answer: $\boxed{12}$

Solution: We first note that 19! has $2^3 * 5^3$ in its prime factorization, which means that $19! \equiv 0 \pmod{1000}$, so $H = 0$. Next, we note that 19! is divisible by 9 and 11, and that the sum of the digits of the provided number is $T + M + H + 33 = T + M + 33$. We also can compute the alternating sum of its digits $(1 - 2 + 1 - \cdots)$to be $T - M - H - 7 = T - M - 7$. This suggests we can use their divisibility rules to write the following.

$$\begin{aligned} T + M + 33 \equiv 0 \pmod 9 &\rightarrow T + M \equiv 3 \pmod 9 \\ T - M - 7 \equiv 0 \pmod{11} &\rightarrow T - M \equiv 7 \pmod{11} \end{aligned}$$

We note that $T + M = 3$ is impossible as the only possible pairs of T and M do not satisfy $T - M \equiv 7 \pmod 9$. We check $T + M = 12$, and find that $T = 4$ and $M = 8$ work for both congruences. Hence, $T + M + H = 4 + 8 + 0 = \boxed{12}$.

Problem 6

The number n can be written in base 14 as $\underline{a}\ \underline{b}\ \underline{c}$, can be written in base 15 as $\underline{a}\ \underline{c}\ \underline{b}$, and can be written in base 6 as $\underline{a}\ \underline{c}\ \underline{a}\ \underline{c}$, where $a > 0$. Find the base-10 representation of n. (AIME I 2018)

Answer: $\boxed{925}$

Solution: Converting to base 10, we can write $\overline{abc}_{14} = a * 14^2 + b * 14 + c$. Similarly, $\overline{abc}_{15} = a * 15^2 + c * 15 + b$ and $\overline{acac}_6 = a * 6^3 + c * 6^2 + a * 6 + c$. These describe the same number n, which means we can engineer the following equation.

$$196a + 14b + c = 225a + 15c + b = 222a + 37c$$

From $225a + 15c + b = 222a + 37c$, we find that $3a + b = 22c$, which means that $c = 1$ as $0 < 3a + b < 44$. Hence, $3a + b = 22$.

From $196a + 14b + c = 225a + 15c + b$, we then find that $-29a + 13b - 14c = 0$, and plugging in $c = 1$ results in $-29a + 13b = 14$. As $b = 22 - 3a$ from the prior paragraph, then

$$\begin{aligned} -29a + 13(22 - 3a) = 286 - 68a &= 14 \\ a &= 4. \end{aligned}$$

We can immediately deduce that $b = 10$ (a digit higher than 9 works since b is only explicitly featured in representations with base > 10). Hence, the base 10 representation of n is

$$n = \overline{acac}_6 = a * 6^3 + c * 6^2 + a * 6 + c = 222a + 37c = 222(4) + 37(1) = \boxed{925}.$$

Problem 7

The positive integers N and N^2 both end in the same sequence of four digits $abcd$ when written in base 10, where digit a is not zero. Find the three-digit number abc. (AIME I 2014)

Answer: $\boxed{12}$

Solution: This means that $N \equiv \overline{abcd} \pmod{10^4}$ and $N^2 \equiv \overline{abcd} \pmod{10^4}$. Accordingly,

$$N^2 - N \equiv \overline{abcd} - \overline{abcd} \equiv 0 \pmod{10^4}.$$

Thus, $N(N-1)$ is divisible by $10^4 = 2^4 * 5^4$. We note that N or $N-1$ cannot be divisible by 2 and 5 simultaneously since the other number will end in either 1 or 9, which cannot happen for a number divisible by 2 and 5 (the factors of 2 and 5 would be shared among N and $N-1$). Thus, one number is divisible by $2^4 = 16$ and the other number is divisible by $5^4 = 625$. We note that $N(N-1) = 625 * 624$ would theoretically work as $624 \equiv 0 \pmod{16}$, but the thousands digit would be 0, which doesn't meet the conditions.

We suppose instead that $N \equiv 0 \pmod{16}$, which would suggest that $N - 1 \equiv -1 \equiv 15 \pmod{16}$ and $N - 1 \equiv 0 \pmod{625}$. Hence, we could try $N - 1 = 625 * 15 \equiv 1 * 15 \equiv 15$

(mod 16), as $625 \equiv 1 \pmod{16}$. This means that $N - 1 = 625 * 15 = 9375$ is a viable solution, so $N = \boxed{9376}$.

Remark: Solving the system of two modular congruences $N - 1 \equiv 0 \pmod{625}$ *and* $N - 1 \equiv 15 \pmod{16}$ *is related to Chinese Remainder Theorem.*

Problem 8

Oscar buys 13 pencils and 3 erasers for 1.00. A pencil costs more than an eraser, and both items cost a whole number of cents. What is the total cost, in cents, of one pencil and one eraser? (AMC 12A 2006)

Answer: $\boxed{10}$

Solution: Let p be the cost of a pencil and e be the cost of an eraser, both in cents. Note that $p > e$. We have $13p + 3e = 100$, which means that

$$13p + 3e \equiv 1p + 0 \equiv p \equiv 1 \pmod{3}.$$

The last part is because $100 \equiv 1 \pmod{3}$. Thus, we can try $p = 1$, $p = 4$, $p = 7$, and so forth. With $p = 1$, we get $e = 29$, which is implausible as $p > e$. With $p = 4$, we get a similar result. With $p = 7$, we get $e = 3$, which aligns with $p > e$. Thus, $p + e = \boxed{10}$.

Problem 9

A palindrome between 1000 and 10, 000 is chosen at random. What is the probability that it is divisible by 7? (AMC 10B 2010)

Answer: $\boxed{\frac{1}{5}}$

Solution: The four-digit palindrome will have the form $\overline{abba}$ for digits a and b. The divisibility rule for 7 states that if $\overline{abb} - 2 * a \equiv 0 \pmod{7}$, then the number $\overline{abba}$ is divisible by 7. More elaborately, we take the last digit of the number, double it, and subtract it from the rest of the number; then, we observe if that result is divisible by 7. Hence,

$$\overline{abb} - 2 * a \equiv 100a + 10b + b - 2a \equiv 98a + 11b \equiv 0 \pmod{7}.$$

Since $98 \equiv 0 \pmod{7}$ and $11 \equiv 4 \pmod{7}$, then $4b \equiv 0 \pmod{7}$. This means that $b \equiv 0 \pmod{7}$, which is the only prerequisite for the palindrome being divisible by 7. Thus, b must be either 0 or 2, and because b is chosen from 0 to 9, the probability is this is equal to $\frac{2}{10} = \boxed{\frac{1}{5}}$.

Problem 10

Arithmetic sequences (a_n) and (b_n) have integer terms with $a_1 = b_1 = 1 < a_2 \leq b_2$ and $a_n b_n = 2010$ for some n. What is the largest possible value of n? (AMC 12A 2010)

Answer: $\boxed{8}$

Solution: Let x be the common difference of the sequence (a_n) and y be the common difference of the sequence (b_n). This means that the nth term of each sequence can be encoded as

$$a_n = (n-1)x + 1 \equiv 1 \pmod{n-1}$$
$$b_n = (n-1)y + 1 \equiv 1 \pmod{n-1}$$

Evidently, $a_n \equiv b_n \equiv 1 \pmod{n-1}$, which means that $a_n - 1$ and $b_n - 1$ are both divisible by $n-1$. To maximize n, we effectively want to find the largest $n-1$ that divides both $a_n - 1$ and $b_n - 1$; in other words, the GCD of $a_n - 1$ and $b_n - 1$.

We know that $a_n b_n = 2010 = 2 * 3 * 5 * 67$. As $1 < a_n \leq b_n$, then we consider each possible value of a_n, which can take on 2, 3, 5, $2*3$, $2*5$, $3*5$, or $2*3*5$. We find that $a_n = 15$ and the corresponding $b_n = 2 * 67 = 134$ results in the largest $\gcd(15-1, 134-1) = 7$, so $n - 1 = 7$ and $n = \boxed{8}$.

Week 3 Homework: Number Theory

Problem 1

Prove that for any four distinct and positive integers, there are two of them, say a and b, such that $a^2 - b^2$ is divisible by 5.

Hint: Consider all the possible values for $n^2 \pmod 5$ *for an integer* n*. This is similar to problem* 1 *in the lecture.*

Problem 2

Suppose that $3x \equiv 2 \pmod 5$. Find the remainder when x is divided by 5; in other words, compute $x \pmod 5$.

Problem 3

Three pirates consult a magical oracle to discover the pass-code N to open a treasure chest. They know that N is a positive integer. The oracle returns three statements:

- When N is divided by 3, it leaves behind a remainder of 2.
- When N is divided by 5, it leaves behind a remainder of 4.
- When N is divided by 7, it leaves behind a remainder of 1.

Exactly one of these statements is a lie and the rest are true. What is the smallest possible value of N?

Problem 4

A toy company can ship toys in boxes that contain 5 toys each. The company plans to deliver $N^2 + 5N + 4$ toys, where N is an integer and $1 \leq N \leq 50$, in boxes to a local neighborhood. Compute the number of possible N such that no toys will be leftover after packing as many complete boxes of toys as possible. *Hint: Factor* $N^2 + 5N + 4$.

Problem 5

Let a_n represent the number a written in base n. For example, 101_2 is the binary number 101. Simplify the following expression into a single base 10 number:

$$111_2 + 222_3 + 333_4 + \cdots + 888_9 + 999_{10} + AAA_{11} + BBB_{12} + \cdots + FFF_{16}.$$

Here, A represents the digit 10, B represents the digit of 11, and, in general, the ith letter of the English alphabet represents the digit of $9 + i$.

Hint: Can one write a general formula for $\overline{XXX}_{X+1}$, which is the three-digit number $\overline{XXX}$ in base $X + 1$?

Problem 6

Isabelle selects a random integer A from the set $\{1, 2, 3, \cdots, 9\}$ and Sonya selects a random integer B from the set $\{0, 1, 2, 3, \cdots, 9\}$. They construct the 5-digit number $\overline{ABABA}$. What is the probability that this number is divisible by 11?

Hint: Consider the divisibility rule for 11.

Week 3 Homework SOLUTIONS

Problem 1

Prove that for any four distinct and positive integers, there are two of them, say a and b, such that $a^2 - b^2$ is divisible by 5.

Hint: Consider all the possible values for $n^2 \pmod 5$ *for an integer* n*. This is similar to problem* 1 *in the lecture.*

Answer: Proof below

Solution: We note that $0^2 \equiv 0 \pmod 5$, $1^2 \equiv 1 \pmod 5$, $2^2 \equiv 4 \pmod 5$, $3^2 \equiv 4 \pmod 5$, and $4^2 \equiv 1 \pmod 5$. Thus, for some integer n, where n is either 0, 1, 2, 3, or 4 $\pmod 5$, then $n^2 \pmod 5$ can assume only three possible values: $0, 1, 4$. If we have four distinct integers, by the *Pigeonhole Principle*, at least two of them, say a and b, when squared, have the same value modulo 5. In other words, $a^2 \equiv b^2 \pmod 5$, which means that $a^2 - b^2 \equiv 0 \pmod 5$, finishing the proof.

Problem 2

Suppose that $3x \equiv 2 \pmod 5$. Find the remainder when x is divided by 5; in other words, compute $x \pmod 5$.

Answer: $\boxed{4}$

Solution: We peruse each of the possible values of $x \pmod 5$: $0, 1, 2, 3, 4$. We find that $3(0) \equiv 0 \pmod 5$, $3(1) \equiv 3 \pmod 5$, $3(2) \equiv 1 \pmod 5$, $3(3) \equiv 4 \pmod 5$, and $3(4) \equiv 2 \pmod 5$. Hence, the only possible answer is $x \equiv 4 \pmod 5$, which implies that the remainder when x is divided by 5 is equal to $\boxed{4}$.

Problem 3

Three pirates consult a magical oracle to discover the pass-code N to open a treasure chest. They know that N is a positive integer. The oracle returns three statements:

- When N is divided by 3, it leaves behind a remainder of 2.
- When N is divided by 5, it leaves behind a remainder of 4.

- When N is divided by 7, it leaves behind a remainder of 1.

Exactly one of these statements is a lie and the rest are true. What is the smallest possible value of N?

Answer: $\boxed{8}$

Solution: We note that the statements are equivalent in modular arithmetic to $N \equiv 2 \pmod 3$, $N \equiv 4 \pmod 5$, and $N \equiv 1 \pmod 7$. We build casework based on which statement is the lie.

- *Case 1:* $N \equiv 2 \pmod 3$ *is the lie.* This means that $N \equiv 4 \pmod 5$ and $N \equiv 1 \pmod 7$, which means we can write

$$5a + 4 = 7b + 1 \to b = \frac{5a+3}{7}$$

  We find that $a = 5$, which means $b = 4$ for minimal values of $\{a, b\}$. Hence, $N \equiv 5a + 4 \pmod{5 * 7} \equiv 29 \pmod{35}$ via Chinese Remainder Theorem.

- *Case 2:* $N \equiv 4 \pmod 5$ *is the lie.* This implies that $N \equiv 2 \pmod 3$ and $N \equiv 1 \pmod 7$. Hence,

$$3a + 2 = 7b + 1 \equiv b = \frac{3a+1}{7}$$

  We find $a = 2$ so $b = 1$, and thus $N \equiv 3a + 2 \equiv 8 \pmod{21}$

- *Case 3:* $N \equiv 1 \pmod 7$ *is the lie.* The truths must be $N \equiv 2 \pmod 3$ and $N \equiv 4 \pmod 5$, so

$$3a + 2 = 5b + 4 \equiv b = \frac{3a-2}{5}.$$

  This means that $a = 4$ so $b = 2$, and thus $N \equiv 3a + 2 \equiv 14 \pmod{15}$.

The smallest possible values in each case are 29, 8, and 14, respectively, so the smallest value is $N = \boxed{8}$.

Problem 4

A toy company can ship toys in boxes that contain 5 toys each. The company plans to deliver $N^2 + 5N + 4$ toys, where N is an integer and $1 \leq N \leq 50$, in boxes to a local neighborhood. Compute the number of possible N such that no toys will be leftover after packing as many complete boxes of toys as possible. *Hint: Factor* $N^2 + 5N + 4$.

Answer: $\boxed{20}$

Solution: We factor $N^2 + 5N + 4 = (N + 4)(N + 1)$, and note that either $N + 4$ or $N + 1$ must have a factor of 5. In other words, either $N + 4 \equiv 0 \pmod 5$ or $N + 1 \equiv 0 \pmod 5$, which translates to $N \equiv 1 \pmod 5$ or $N \equiv 4 \pmod 5$. Note that N ranges from 1 to 50, so there are 10 numbers $(1, 6, \cdots, 46)$ satisfying $N \equiv 1 \pmod 5$ and 10 numbers $(4, 9, \cdots, 49)$ satisfying $N \equiv 4 \pmod 5$. The total number of possible N is $10 + 10 = \boxed{20}$.

Problem 5

Let a_n represent the number a written in base n. For example, 101_2 is the binary number 101. Simplify the following expression into a single base 10 number:

$$111_2 + 222_3 + 333_4 + \cdots + 888_9 + 999_{10} + AAA_{11} + BBB_{12} + \cdots + FFF_{16}.$$

Here, A represents the digit 10, B represents the digit of 11, and, in general, the ith letter of the English alphabet represents the digit of $9 + i$.

Hint: Can one write a general formula for $\overline{XXX}_{X+1}$, which is the three-digit number $\overline{XXX}$ in base $X + 1$?

Answer: $\boxed{18480}$

Solution: Each term in the sum has the three-digit form $\overline{XXX}_{X+1}$, which we note is 1 less than the four-digit number $\overline{1000}_{X+1}$. This is because $\overline{XXX}_{X+1}$ is the largest three-digit number we can construct in base $X+1$ and $\overline{1000}_{X+1}$ is the smallest four-digit number. Translating to base 10, we can write

$$\overline{XXX}_{X+1} = \overline{1000}_{X+1} - 1_{X+1} = (X+1)^3 - 1$$

Thus, the sum in the problem can be reconfigured as

$$\begin{aligned} 111_2 + 222_3 + 333_4 + \cdots + 888_9 + 999_{10} + AAA_{11} + BBB_{12} + \cdots + FFF_{16} \\ = (2^3 - 1) + (3^3 - 1) + (4^3 - 1) + \cdots + (16^3 - 1) \\ = (2^3 + 3^3 + 4^3 + \cdots + 16^3) - 15 \\ = (1^3 + 2^3 + 3^3 + 4^3 + \cdots + 16^3) - 1 - 15. \end{aligned}$$

Utilizing the cubic sum formula, this becomes $(\frac{16(16+1)}{2})^2 - 16 = \boxed{18480}$.

Problem 6

Isabelle selects a random integer A from the set $\{1, 2, 3, \cdots, 9\}$ and Sonya selects a random integer B from the set $\{0, 1, 2, 3, \cdots, 9\}$. They construct the 5-digit number $\overline{ABABA}$. What is the probability that this number is divisible by 11?

Hint: Consider the divisibility rule for 11.

Answer: $\boxed{\dfrac{4}{45}}$.

Solution: In order for $\overline{ABABA} \equiv 0 \pmod{11}$, then the alternating sum of its digits must be divisible by 11. In other words,

$$A - B + A - B + A = 3A - 2B \equiv 0 \pmod{11}.$$

The total number of possible $\overline{ABABA}$ numbers is equal to $9 * 10 = 90$, as we have 9 choices for A and 10 choices for 10. We aim to compute how many of these satisfy $3A - 2B \equiv 0 \pmod{11}$, a congruence that beseeches the use of casework.

- *Case 1:* $3A - 2B = -11$ We find $3A = 2B - 11$, so we can try $B = 7$, which gives $A = 1$. This turns out to be the only (A, B) that works here.
- *Case 2:* $3A - 2B = 0$. We find $3A = 2B$, so we can verify the following pairs of (A, B): $(2, 3), (4, 6), (6, 9)$. There are 3 numbers here.
- *Case 3:* $3A - 2B = 11$. For $3A = 11 + 2B$, we can verify the following pairs of (A, B): $(5, 2), (7, 5), (9, 8)$. There are 3 numbers here.
- *Case 4:* $3A - 2B = 22$. We have $3A = 22 + 2B$, and thus we can verify the pairs of (A, B): $(8, 1)$, which turns out to be the exclusive one that works.

Adding up these cases leads to $1+3+3+1 = 8$ numbers divisible by 11. Hence, the probability is equal to $\frac{8}{90} = \boxed{\frac{4}{45}}$.

Week 4: Diophantine Equations Problems

Problem 1

In an ancient civilization, a sword is worth 100 bronze coins and a furnace is worth 270 bronze coins. To resolve debt, a citizen is allowed to pay another citizen another some number of swords and furnaces, while receiving some number of swords and furnaces in exchange. For instance, a citizen could pay a debt of 440 coins to another citizen by giving them two furnaces and receiving a sword. What is the smallest amount of positive debt that can be paid using a transaction of swords and furnaces?

Problem 2

Prove that (a) $\sqrt{2}$ and (b) $\sqrt{3}$ are irrational.

Hint: If a number x is rational, then it can be written as a fraction $\frac{m}{n}$ where $\gcd(m, n) = 1$. Try using a proof by contradiction.

Problem 3

Two different prime numbers between 4 and 18 are chosen. When their sum is subtracted from their product, which of the following numbers could be obtained? (AMC 12 2000)

(A) 21 (B) 60 (C) 119 (D) 180 (E) 231

Problem 4

Each ice cream scoop at the store costs a whole number of cents. If Damian spends all of his money, he can purchase either 15 scoops of chocolate, 16 scoops of pistachio, 20 scoops of strawberry, 14 scoops of banana, and n scoops of caramel. If a scoop of caramel ice cream costs 100 cents, what is the smallest possible cost of a scoop of each flavor (chocolate, pistachio, strawberry, banana, caramel)?

Problem 5

Let $P(x)$ be a polynomial with integer coefficients that satisfies $P(17) = 10$ and $P(24) = 17$. Given that $P(n) = n + 3$ has two distinct integer solutions n_1 and n_2, find the product $n_1 \cdot n_2$. (AIME II 2005)

Problem 6

Let A, M, and C be digits with

$$(100A + 10M + C)(A + M + C) = 2005$$

What is A? (AMC 12A 2005)

Problem 7

A park in Alaska has three reindeer for each bear, three birds for each goose, and five wolves for each reindeer. Which of the following numbers cannot be the total number of reindeer, bears, birds, geese, and wolves in the wildlife park?

(A) 46 **(B)** 53 **(C)** 55 **(D)** 58 **(E)** 60

Problem 8

Suppose that in an American football game between the Kansas City Chiefs and Tampa Bay Buccaneers, there are no 2-point safeties and after each touchdown, an extra point kick is attempted and each one is successful. This essentially means the only scores are a field goal for 3 points and a touchdown for 7 points. What is the greatest score that cannot be achieved by a team in this game?

Problem 9

Find $3x^2y^2$ if x and y are integers such that $y^2 + 3x^2y^2 = 30x^2 + 517$. (AIME 1987)

Problem 10

Positive integers a and b satisfy the condition

$$\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0.$$

Find the sum of all possible values of $a + b$. (AIME II 2013)